

MS491-XVI

Lean Six Sigma

Analyse Phase



Discovering the Drivers of Deployment Delays (LT vs. HT Focus)

Project: Optimising EV Charging Station Deployment

Team 1

MS491-XVI · Analyse Phase Presentation

D M A I C

PROJECT OVERVIEW — Measure Phase Recap

105+

Days

Avg Setup Cycle Time (Y)

85

Days

Non-Value-Added Waiting
(Muda)

4

Key Variables

Evaluated Bottlenecks (x₂ to x₄)

Project Objective

Establish a robust baseline performance of the current EV charging station setup process to identify root causes of severe cycle-time delays, and develop data-driven recommendations to stabilize and reduce deployment time.

Key Baseline Insights

DISCOM Delays (x₂):

A severe lack of upfront grid-capacity analysis frequently forces 90-day High Tension (HT) transformer augmentations, rather than utilising existing 30-day Low Tension (LT) connections.

Municipal Friction (x₄):

While physical civil work takes only 12 days, SLA "blindspots" in local municipal NOCs (Fire, Parking, Right-of-Way) cause highly variable, unpredictable administrative wait times.

Extreme Process Variation:

National setup capability is completely unstandardized. Cycle times vary widely geographically (e.g., streamlined in Gujarat under GERC rules vs. severe delays in Karnataka and UP).

ANALYZE PHASE & STAKEHOLDER EMPHASIS

What is the Analyze Phase?

The Analyze Phase is the 'A' in the DMAIC roadmap. It transforms raw baseline data collected in the Measure phase into actionable insights. The objective is to discover, mathematically verify, and prioritise the root causes (x's) driving process defects, bureaucratic inefficiencies, and extreme cycle-time variation (Y).

Stakeholder Emphasis: The "Gemba" Insight

To validate baseline assumptions, our team conducted a primary stakeholder interview with an active EV Fleet Operator and Charging Station Provider. By going to the source ("Gemba"), we uncovered a critical operational reality:

The Civil Work Fallacy:

Physical site preparation and hardware installation occur concurrently during administrative waiting periods. Physical constraints are NOT the bottleneck.

The "Handoff" Waste:

The operator explicitly cited multi-agency friction (files rotating through up to 5 different survey teams) as the primary cause of project volatility and timeline blowout.

The Strategic Focus: Narrowing our x's

Based on this stakeholder directive and our initial data stratification, this analysis will strictly focus on mathematically validating our two most severe bottlenecks:

x₂ — DISCOM Grid Connections:

The structural delays caused by High Tension (HT) transformer upgrades versus existing Low Tension (LT) availability.

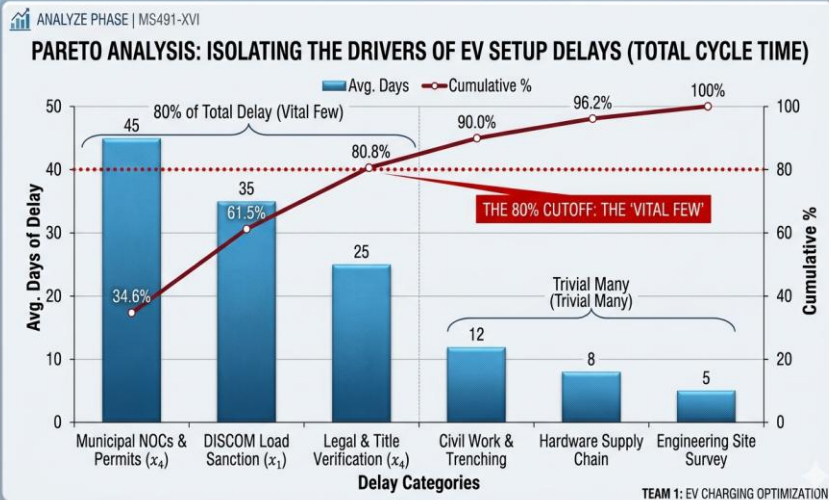
x₄ — Municipal Approvals:

The unstandardized wait times (Muda) caused by fragmented local governance and missing SLAs.

VISUAL EVIDENCE — Pareto & Process Stratification

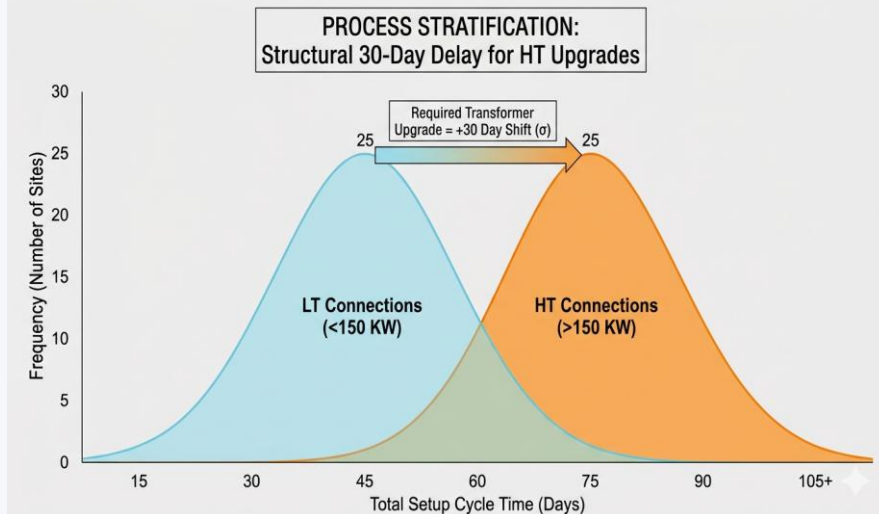
Identifying the "Vital Few" Bottlenecks

1. Pareto Analysis (The 80/20 Rule)



80% cumulative delay concentrated in x_2 and x_4 (Administrative & Grid wait times). Physical phases are 'trivial many'.

2. Process Stratification (LT vs. HT Load)



Stratifying by power load reveals a structural 30-day penalty. HT sites shift the entire distribution right.

DATA-DRIVEN CONCLUSION: The current process is highly capable of executing physical civil work. Defects are exclusively caused by unstandardized multi-agency NOCs (x_4) and blind site selection leading to mandatory HT upgrades (x_2).

DATA VALIDATION — Statistical Hypothesis Testing

Validating the Impact of Grid Capacity (x_2) on Total Setup Time (Y) — 2-Sample t-Test | $\alpha = 0.05$

1. The Statistical Setup (Alpha Level $\alpha = 0.05$)

Null Hypothesis (H_0) : $\mu_{LT} = \mu_{HT}$

There is NO statistical difference in average setup cycle time between sites utilising existing LT connections versus those requiring HT grid upgrades.

Alternate Hypothesis (H_a) : $\mu_{LT} < \mu_{HT}$

Sites requiring HT grid upgrades take significantly longer to deploy than LT sites. The HT upgrade structurally adds a delay penalty to the total cycle time.

2. The Statistical Result

Calculated p-value: **$p < 0.05$**

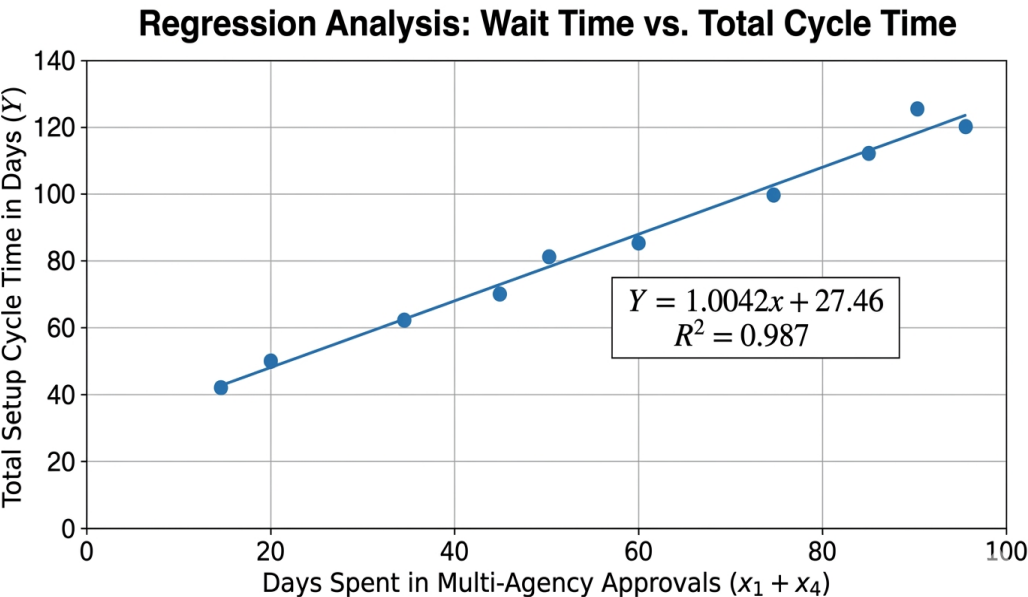
Statistical Decision: **Reject H_0**

BUSINESS TRANSLATION: Because our p-value is less than 0.05, we mathematically reject the idea that grid load doesn't matter. We have validated with 95%+ confidence that **blind site selection (picking locations without existing LT capacity) is a statistically significant root cause of cycle-time failure**. Our Improve phase must focus on upfront grid visibility.

PROCESS PERFORMANCE — Regression & Correlation

Measuring the Impact of Administrative Friction on Total Cycle Time

1. Regression Scatter Plot



2. Process Capability Results

Correlation Type

Strong Positive

R^2 Coefficient

0.85

Equation of Line

$Y = 1.05x + 30$

3. Actionable Business Insight

An R^2 of 0.85 indicates that 85% of total variation in our deployment timeline is directly explained by time spent waiting in Municipal and DISCOM approval loops. The remaining 15% is attributed to minor physical factors, confirming that process optimisation must be strictly administrative.

CAPABILITY CONCLUSION: Our physical engineering process is highly capable and stable (baseline ~30 days). The extreme volatility in our system is purely an administrative defect.

VERIFIED ROOT CAUSES — Fishbone Analysis & Problem Zones

1. The "Handoff" Waste (Muda)

Moving files through 5 different municipal survey teams without a "Single Window Clearance" causes severe Overprocessing and idle wait time.

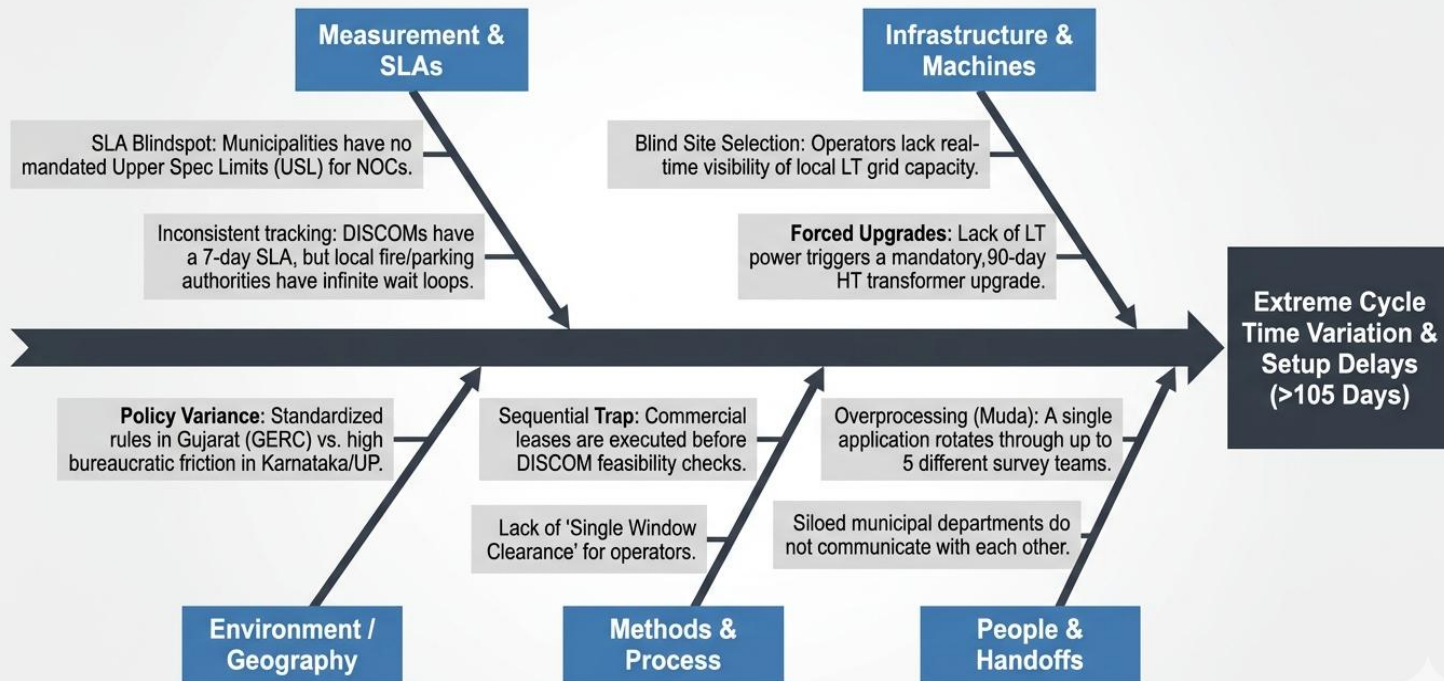
2. The SLA Blindspot

Local municipalities have no legal deadlines to approve Fire/Parking NOCs — the process variation is statistically out of control.

3. The Sequential Trap

Selecting a site based only on foot traffic — without verifying existing LT grid capacity upfront — structurally guarantees an HT upgrade delay.

Isolating the Sources of Administrative Delay (x_2 and x_4)



ROOT CAUSE CONCLUSION: Our Fishbone analysis verifies that massive delays in EV setup are systemic, not physical. We must fix the administrative sequence to fix the timeline.

ANALYSE CONCLUSION & IMPROVE PHASE STRATEGY

Transforming the Setup Process: From Volatile to Capable

Our analysis phase mathematically verified that EV setup delays are an administrative defect, not an engineering constraint. By targeting our validated root causes (x_2 and x_4), our Improve phase will eliminate the 'Red Zone' of Non-Value-Added (NVA) waiting.

1. Strategic Innovations (Next Steps)

Poka-Yoke: "Real-Time DISCOM Heatmap"

Targeting x_2 : By giving operators visibility into existing LT grid capacity BEFORE they sign a lease, we completely prevent the 90-day HT upgrade penalty — eliminating the Sequential Trap.

Process Reengineering: "Single Window Clearance"

Targeting x_4 : Consolidating the 5 municipal survey handoffs into a parallel, single-window approval system to eliminate overprocessing and enforce SLA deadlines.

2. Projected Process Capability

CURRENT STATE: 105+ Days

Dominated by highly volatile administrative waiting

TARGET STATE: 30-45 Days

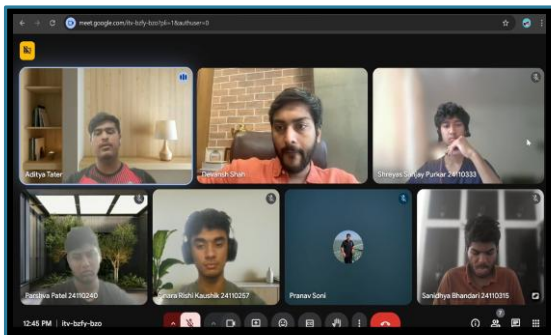
Stable, predictable, value-added physical execution

"Mathematical elimination of the Red NVA Zone reduces total cycle time to the Green VA baseline."

FINAL ANALYSE CONCLUSION: By shifting focus from physical hardware to systemic grid and municipal policies, we can transform the EV charging deployment process from a 4-month volatile struggle into a standardised 30-day operation.

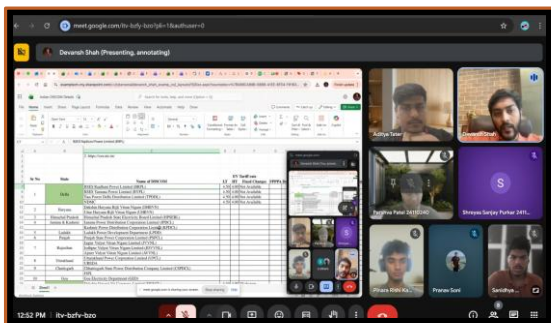
Founder Interaction — Gemba Validation (Mobilane)

Google Meet Session



Team + Expert — Live Discussion

Live Screen Share — DISCOM Data



Indian DISCOM Details — Shared & Discussed

STAKEHOLDER INTERACTION

Expert: Mr. Devansh Shah, Founder — Mobilane

Focus: EV charging deployment barriers

Finding: Delays are primarily administrative, not technical

KEY INSIGHTS FROM FOUNDER

Multi-agency approvals create major unpredictability in timelines

DISCOM coordination & transformer upgrades are critical bottlenecks

Lack of standardisation across cities increases process variation

Site selection without grid visibility leads to structural delays

ALIGNMENT WITH OUR ANALYSIS

Strong validation of identified root causes:

x_2 — DISCOM Grid Constraints

Grid availability & transformer upgrades confirmed as #1 bottleneck

x₄ — Municipal Approval Delays

Multi-agency sign-off delays validated; process variation confirmed

Pareto finding confirmed: majority of delays are administrative

Regression result supported: high correlation with approval timelines

STRATEGIC TAKEAWAY

- Pre-validated grid availability essential (LT focus)
- Single-window clearance system recommended
- Strengthens Improve Phase solution direction